Cybersecurity Incident Report: Network Traffic Analysis

Part 1: Provide a summary of the problem found in the tcpdump log
From my analysis of the tcpdump logs, I identified a significant issue with DNS communication. The client system repeatedly attempted to send DNS queries over UDP port 53 to resolve the domain name yummyrecipesforme.com. These requests were visible as outgoing packets from the client to the DNS server. However, instead of receiving valid DNS responses, the client consistently received ICMP messages indicating "UDP port 53 unreachable." This shows that the destination server was not accepting traffic on the DNS port. The repeated pattern of DNS requests followed by ICMP error responses confirms that the DNS resolution process was failing. As a result, the client system was unable to obtain the IP address associated with the domain, which prevented access to the website. In summary, the tcpdump logs reveal that DNS queries were unsuccessful due to an unreachable port 53, as confirmed by ICMP error messages, leading to a failure in domain name resolution.

Part 2: Explain your analysis of the data and provide at least one cause of the
From my analysis of the tcpdump data, I observed that the client system was repeatedly sending DNS queries over UDP port 53 in an attempt to resolve the domain name yummyrecipesforme.com. These requests indicate normal DNS activity where the system is trying to translate a domain name into an IP address.

However, the logs show that instead of receiving valid DNS responses, the client consistently received ICMP error messages stating “UDP port 53 unreachable.” This indicates that the destination server was not accepting traffic on the DNS port.

The repeated sequence of DNS requests followed by ICMP error responses confirms that the DNS resolution process was failing. As a result, the client system could not obtain the IP address needed to access the website, which led to the connection failure.

One likely cause of this incident is that the DNS service on the server was not running or misconfigured, meaning no application was listening on port 53. Another possible cause is that a firewall or network security control was blocking UDP traffic on port 53, preventing successful communication.

In conclusion, the data shows that the failure of DNS communication, confirmed by ICMP “port unreachable” messages, was the main cause of the incident.